Project Scope

Plant Disease Detection System

1. Introduction

Plant diseases significantly affect crop productivity and food supply. Early detection of plant diseases helps farmers take timely action to prevent crop damage. The **Plant Disease Detection System** uses image processing and machine learning to identify diseases in plant leaves.

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2. Project Objective

The main objective of this project is to develop a system that can **detect plant diseases automatically by analyzing leaf images**.

The system aims to:

- Identify plant diseases at an early stage.
- Help farmers take quick preventive measures.
- Reduce crop loss and improve agricultural productivity.

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3. Scope of the Project

The scope of the Plant Disease Detection System includes the following:

1. Image Input

- Users can upload plant leaf images through the system.

2. Image Processing

- The system processes the image using computer vision techniques.

3. Disease Detection

- Machine learning algorithms analyze the image and detect possible plant diseases.

4. Disease Identification

- The system identifies the specific disease affecting the plant.

5. Result Display

- The system displays:
    - Disease name

    - Disease description
    - Possible treatment suggestions

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4. Key Features

- Upload plant leaf images.
- Automatic disease detection.
- Fast and accurate analysis.
- User-friendly interface.
- Disease information and treatment suggestions.

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5. Technologies Used

- **Python**
- **Machine Learning / Deep Learning**
- **React Native**
- **TensorFlow / Keras**
- **Dataset of plant leaf images**

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6. Target Users

- Farmers
- Agricultural researchers
- Students working on agriculture projects

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7. Limitations

- Detection accuracy depends on image quality.
- The system may support only limited plant species.
- Requires internet or computing resources.

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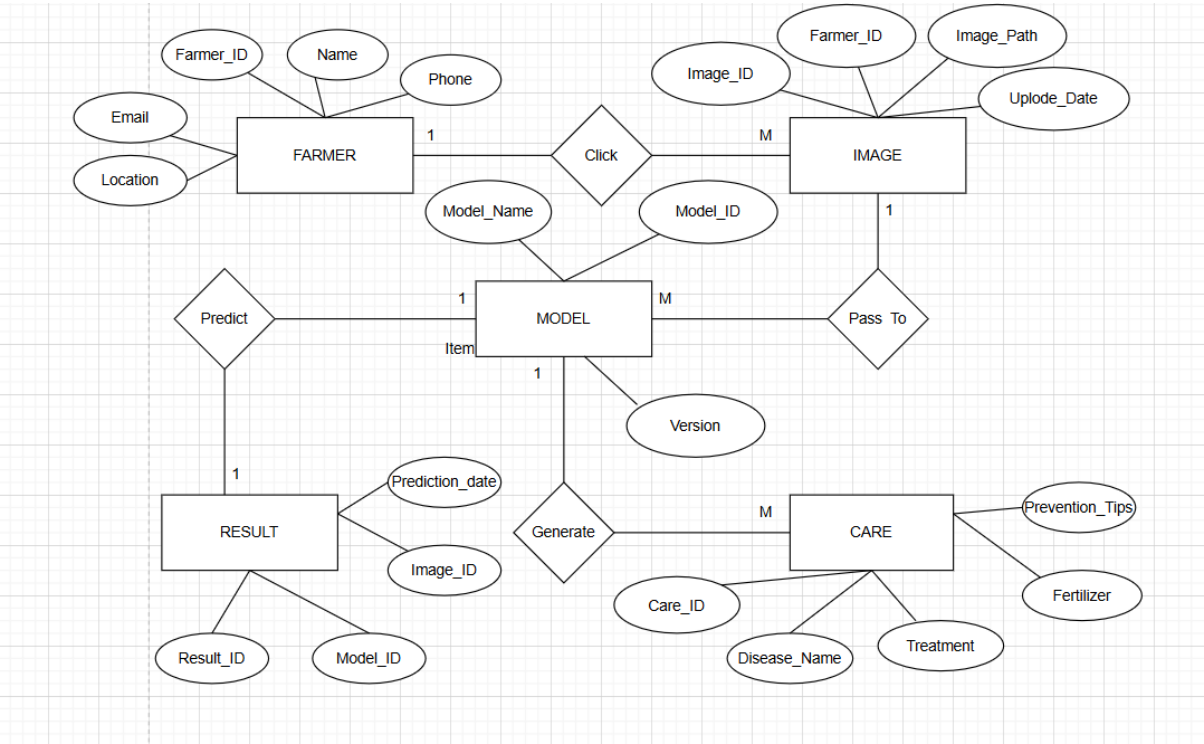
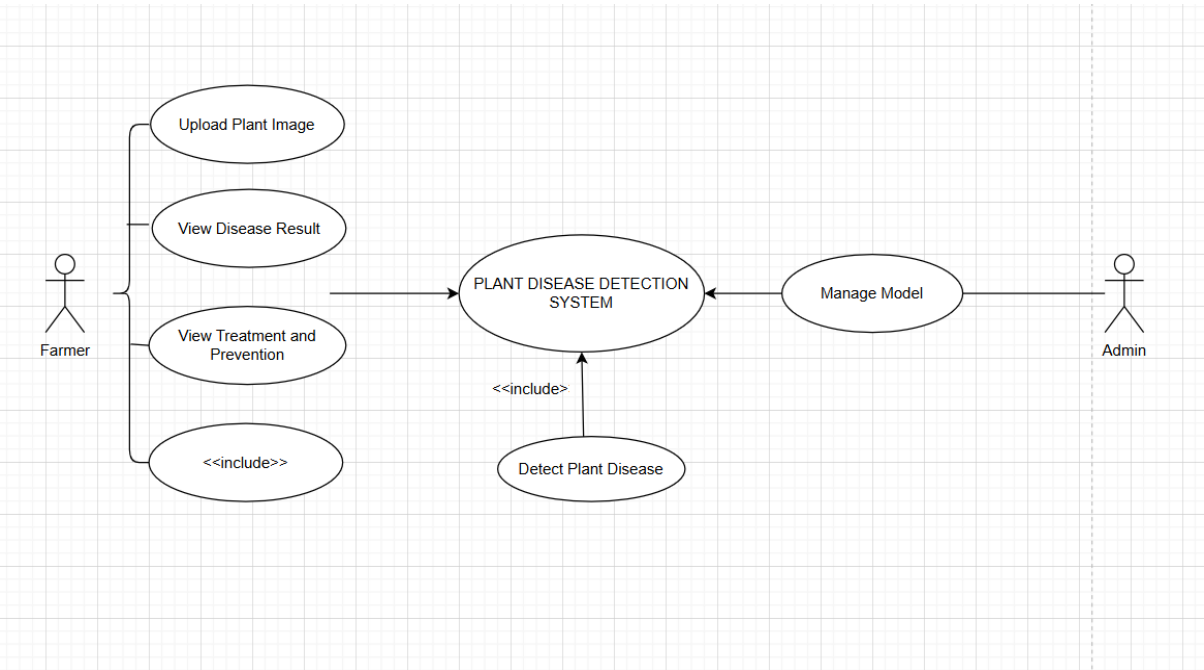
8. Objective

- Early Detection of Plant Diseases
- Improve Crop Health
- Use of Image Processing and Machine Learning
- Reduce Crop Loss
- Provide Disease Information and Solutions
- Create a User-Friendly System

9. Expected Outcome

The system will help in **early disease detection**, reducing crop damage and improving farming efficiency.

ERD(Entity Relationship Diagram)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